CPRG 307 - Database Programming

Description
This course builds on the programming skills students have already learned, with a focus on database-specific structured languages and embedding data requests into these languages. Students will also explore the importance of efficient code creation using an industry-identified database vendor language. Other topics include basic programming structures, advanced concepts in database programming and events.

3 Credits

Time Guidelines
The standard instructional time for this course is 60 hours.

Prerequisites

- CPRG 250

Course Assessment

Learning Activities	40%
Assignments	60%
Total	100%

Use of Turnitin

The instructor may submit student work in this course to Turnitin's text-matching software program, to help assess the academic integrity of student work in this course. Turnitin results may be considered as one piece of evidence in academic misconduct hearings. Turnitin is an American company that keeps no unencrypted student identity data in the United States unless the students themselves choose to share this information in their submission. Students should limit their sharing of personally identifiable information by not including their names and student ID numbers within the text body of submitted assignments. SAIT has carefully reviewed this company's data management procedures.

Other Course Information
Learner Engagement:

In order to be successful, the learner is expected to be engaged in learning activities for a total of 9 to 12 learning hours per course per week, which includes both in-class and out-of-class time.

Course Learning Outcomes
1. Describe Database Programming Languages

Objectives:

1.1 Describe how logic can be executed within a database system in addition to data storage.
1.2 Explain the purpose of grouping data related logic into executable units.
1.3 Distinguish between temporary execution contexts and reusable database resident logic at a conceptual level.
1.4 Explain how executable logic interacts with stored data during execution.
1.5 Recognize that database logic can be developed incrementally before being formalized into persistent structures.
1.6 Describe database managed mechanisms that supply ordered or unique values independently of application logic.
1.7 Explain how database systems describe and expose information about their own structure and behavior through metadata.

2. Apply the basic structure of a database programming language.

Objectives:

2.1 Describe the structure of a programming block.
2.2 Describe data declarations for variables including variables that dynamically retrieve data.
2.3 Explain the use of built-in functions and/or procedures within database programming.
2.4 Employ commands in a database programming block to manipulate and retrieve data.

3. Use database programming looping structures and conditionals to manipulate and retrieve data.

Objectives:

3.1 Explain how looping logic enables repeated execution in response to data conditions.
3.2 Explain how conditional logic controls execution flow based on data conditions.
3.3 Apply looping and conditional logic to control execution flow in response to data conditions.

4. Define the available techniques used to manage database runtime errors

Objectives:

4.1 Describe a database vendor's error-handling environment.
4.2 Demonstrate the integration of error handlers into a program.
4.3 Use predefined error handling provided by a database vendor.
4.4 Use user-defined error handling.
4.5 Apply error codes and messages retrieved from the built-in functions

5. Use events with a database.

Objectives:

5.1 Describe the concepts and structures of events.
5.2 Demonstrate the process of creating events.
5.3 Explain how data driven events differ from logic explicitly invoked by users or applications.
5.4 Identify appropriate and inappropriate use cases for data driven events within a database environment.
5.5 Utilize commands to modify, enable, and disable events within a database.

SAIT Policies and Procedures:
Students should familiarize themselves with SAIT's Policies and Procedures, including but not limited to those in the Academic Student, Conduct, and Health, Safety, and Environment domains.

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